On Modality Awareness in Vision-Language Models

Desmond P. Hartigan-O'Connor, Ayda Sultan, Achintyan Kaliraj, Justin Zhang, and Ryan Xu

Latent Visual Reasoning: Perception, Imagination, and Multimodal Thought
Workshop at the 2027 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)
Accepted for presentation

Abstract

Vision-language models (VLMs) fuse text and image activations into a shared residual stream to achieve high-quality performance in multimodal tasks, but it remains poorly understood how these models distinguish tokens by modality and how modality affects the model's output. We propose that modality is a complex concept, requiring multiple dimensions to encode. We compute the difference between the mean text and image activations, showing that a collection of these difference-in-mean vectors encode modality and causally affect model output. We observe the impact of this modality subspace across four open-source VLMs from two model families. Ablating modality in the model's activations decreases the performance of linear probes trained to differentiate between text and image tokens and reduces the model's preference for utilizing text information when generating output, while steering along the modality vector alters the model's attention. However, hundreds of linear probe directions must be removed before modality is fully undecodable, suggesting that modality is a complex concept that requires many dimensions to fully capture. Our findings motivate further investigation into the underlying methods in which VLMs encode and react to modality information.